UniMentor AI

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             |             EMPATHY MAP              |
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        +---------------+     +-----------------+
        |     SAYS      |     |     THINKS      |
        +---------------+     +-----------------+
        | "I lose track |     | "I need better |
        |  of deadlines"|     |  structure"      |
        | "I want clear |     | "Which clubs     |
        |  feedback"    |     |  matter to me?" |
        +---------------+     +-----------------+

+---------------+                         +-----------------+
|     DOES      |                         |     FEELS       |
+---------------+                         +-----------------+
| Juggles apps  |                         | Overwhelmed     |
| Random events |                         | Disconnected    |
| Limited study |                         | Anxious         |
+---------------+                         +-----------------+

        +---------------+     +-----------------+
        |     PAINS     |     |     GAINS       |
        +---------------+     +-----------------+
        | Disorganized  |     | Unified system |
        | Slow feedback |     | Personalised    |
        | Hard to find  |     | feedback +      |
        | right events  |     | suggestions      |
        +---------------+     +-----------------+
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Evidence Synthesis

Evidence Snapshot – Quotes (Academic + Engagement)

Theme	Representative Quote
Feedback & Clarity	“I rarely get feedback when I need it, so I don’t know how to improve.”
Organization	“My deadlines and events are scattered everywhere—I can’t track anything.”
Engagement Fit	“Most events I hear about aren’t relevant to what I actually like.”
Schedule Overlap	“Things keep clashing… I wish my calendar just organized itself.”

Pattern Synthesis

- Students struggle with **fragmented academic + extracurricular information**.
- They desire **personalized structure and relevance** across their university life.

Insight Summary

Students want a single intelligent system that brings clarity, relevance, and coordination across academics and campus engagement.

JTBD + HMW

Final Job-To-Be-Done (JTBD)

When managing my academic workload and campus life, I want a single intelligent system that organizes my schedule, provides timely learning support, and recommends relevant activities, so I can stay on track, reduce stress, and make the most of my university experience.

Prioritized HMW Statements

HMW 1:

How might we help students centralize and structure their academic and extracurricular information so they always know what to do next?

HMW 2:

How might we personalize feedback, recommendations, and event discovery so each student feels guided and connected to the right academic and social opportunities?

Low-Fi Prototype

Concept: AI Student Companion (Academic + Engagement Layers)

A simple, unified interface where students see their **daily schedule**, receive **AI-generated recommendations**, and get **automated academic feedback**.

Core Functions in Prototype -

- **Smart Schedule:** Merges classes, deadlines, club events into one view.
- **AI Feedback:** Instant insights on quizzes/assignments.
- **Recommendations:** Clubs, events, and study plans based on interests + workload.
- **One-Click Actions:** Join event, add to calendar, preview workload impact.

Low-Fi Sketch -

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+----------------------------------------------------+
|           AI STUDENT COMPANION               |
+----------------------------------------------------+
| TODAY                                    |
| [09:00] Data Analytics Class                  |
| [11:00] Assignment Reminder: "Submit Case Study"   |
| [16:00] Club Event: "Photography Meetup" (JOIN)    |
+----------------------------------------------------+
| AI RECOMMENDS                            |
| • Study Plan: 45 min review for tomorrow's quiz    |
| • Event Fit: "Cultural Night" matches your interests |
+----------------------------------------------------+
| QUICK FEEDBACK                           |
| Your last quiz: "You struggled with Topic 3.       |
| Suggested recap videos →"                  |
+----------------------------------------------------+
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Purpose of Prototype

Visualizes the **core value**: one place for **organization, support, and personalized engagement**.

Initial KPI + Decision Threshold

Objective of Early Test

Validate whether the students choose AI Companion App to meaningfully improve **student organization**, **clarity**, and **engagement relevance**.

Primary KPI (v1 Test)

KPI: *Student downloads and uses the app*

Metric: Number of 1st year students downloading the AI companion App

Measurement Method: Number of downloads in Apple and Android playstore

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Secondary KPIs

- **%of students seeing improvement in speed of feedback and better management of activities and feel clarity**
- **Recommendation Relevance:** % of recommended events or study tasks students mark as "useful."

Decision Threshold (Go / No-Go)

Go: If **≥ 60%** of students download and use the app

Review/Pivot: If **40–59%** download and use the app

No-Go: If **< 40%** download and use the app

Why These KPIs Matter

They test whether the core value proposition—**adoption** + **centralization + personalization**—encourages students to download and use the app

NaBC: Need + Approach

N — Need (What Students Struggle With)

University students face *fragmented academic and campus life management*. They juggle: - Scattered deadlines across multiple platforms - Slow or unclear academic feedback - Random, irrelevant event discovery - Overlapping schedules and poor activity planning

Core Need: A unified, intelligent system that brings *clarity, organization, and personalized relevance* across academics and engagement.

A — Approach (Our Solution Concept)

We propose an **AI Student Companion-UniMentor AI** that integrates two layers:

1. Academic Layer

- AI-supported teaching assistance
- Automated assessment + instant feedback
- Personalized learning paths
- Unified timetable + smart academic calendar

2. Engagement Layer

- Student segmentation (interests, courses, nationality, clubs)
- Personalized event recommendations and notifications
- Activity planning + conflict resolution

Value Proposition: One intelligent app that centralizes tasks, improves clarity, and helps students stay academically on track while meaningfully connected to campus life.

B — Benefits per Costs

- **Clear student value:** timetable conflicts reduced, better clarity, and improved academic feedback.
- **Quantifiable impact:**
  – Weekly Active Usage projected ≥ 30%
  – Missed mandatory sessions reduced ≥ 15%
  – Event engagement increased ≥ 20%
- **Operational efficiencies:** less administrative time for faculty, better communication flow across campus.

- **Cost efficiency:** pilot requires minimal infrastructure and low integration cost, making ROI positive even at small scale.

Core Benefit: One unified system replaces multiple disconnected tools students currently juggle.

C — Competition

Current alternatives:

• Generic calendars
• LMS notifications
• Club Instagram pages / email newsletters
• Standalone study apps

Our differentiation:

• Only solution integrating **both academic and engagement layers** in real time
• Personalized recommendations powered by unified data from LMS + clubs + SIS
• Scheduling optimizer that considers academic load + extra-curricular interests
• HITL (Human-in-the-loop) ensures safe AI use in assessment
• Faster, coordinated communication between students, faculty, and clubs

Defensibility:

Integration depth + multi-layer personalization + HITL oversight.

Our Edge

• Only solution combining both layers (academic + engagement)
• Real-time personalization using a unified student profile
• LMS + SIS integration → accurate schedule optimization
• Human-in-the-loop for safe academic assessment

Business Model Canvas

Customer Segments

- First-year students (all degrees)
- Student clubs & campus organizations

Value Proposition

- AI-optimized unified timetable (academic + engagement)
- Faster academic feedback + personalized recommendations
- Relevant event discovery based on interests & workload

Channels

- Mobile app
- LMS integration (Canvas LTI)
- Email + push notifications

Customer Relationships

- Automated assistance + human review for assessments (HITL)

Revenue / Cost Logic

- University license + club promotion fees
- Low operational cost during pilot

Key Resources

- LMS/SIS data access
- AI models + cloud infrastructure

Key Activities

- Data integration, model training, user onboarding

Key Partners

- University IT, Student Affairs, Student Clubs

Cost Structure

- Development, hosting, data processing, support

Test Card

Hypothesis

First-year students who receive a personalized AI timetable will show ≥30% weekly active use and ≥15% reduction in missed activities.

Test

Deploy dual-layer timetable (academic + engagement) to 200 first-year students (100 test / 100 control).

KPI

- Primary KPI: Weekly Active Usage ≥ 30%
- Secondary KPI: Missed sessions reduced ≥ 15%
- Activation KPI: ≥ 50% sign-ups within 14 days

Method

A/B test — usage tracking via app analytics + attendance logs.

Time Window

4 weeks

Risks & Mitigation

- Low opt-in → promotion via faculty & LMS
- Data issues → fallback manual import
- Model errors → HITL review

Decision Threshold

Go ≥ 15% improvement | Pivot 5–15% | No-Go < 5%

Technical Approach & Architecture

Data Sources

• LMS (Canvas): classes, deadlines, grades
• SIS: enrollments, timetables
• Clubs DB: events & memberships
• University calendar

Processing & Infrastructure

• ETL pipelines → central database (PostgreSQL)
• Cloud model hosting (Python/FastAPI)
• Real-time notification service

AI Components

• Scheduling Optimizer (constraints + preferences)
• Recommendation Engine (events + study tasks)
• Assessment Feedback Generator (NLP + rubric alignment)

Human-in-the-Loop

• Instructor dashboard for reviewing and overriding AI-generated academic feedback

Security & Governance

• Role-based access, encryption, audit logs, PII minimization

KPIs & Measurement Strategy

Pilot Metrics — What Success Looks Like

Primary KPI

- Weekly Active Usage: ≥30%

Secondary KPIs

- Missed mandatory sessions reduced ≥15%
- Recommendation relevance ≥ 20% “useful” votes
- Assessment agreement (AI vs human) ≥ 80%

Activation Metrics

- ≥50% sign-up rate in first 2 weeks
- ≥3 weekly interactions per user (calendar, events, reminders)

Measurement Methods

- App analytics dashboard
- Attendance logs from SIS
- In-app feedback surveys
- Control vs treatment group comparison

Adoption Plan & Governance

Adoption Strategy (0–30 Days)

• Launch through LMS & orientation week
• Student ambassadors for each faculty
• Two 30-minute micro-training sessions for instructors
• Weekly feedback check-ins

Governance & Data Protection

• Transparent opt-in for data usage
• PII minimization + hashed identifiers
• Logged AI decisions + human overrides
• Monthly bias + accuracy review
• Clear escalation path for feedback disputes

Why it Matters

Builds trust, reduces resistance, supports ethical AI adoption.